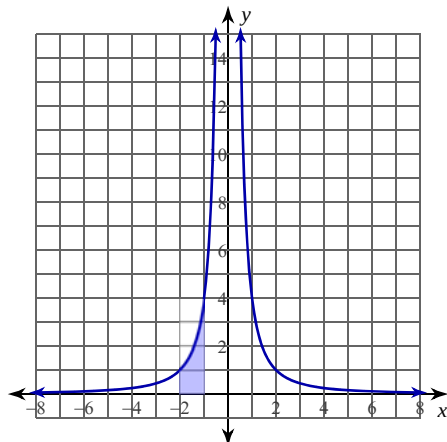


Area Under a Curve

Date _____ Period _____

For each problem, find the area under the curve over the given interval.

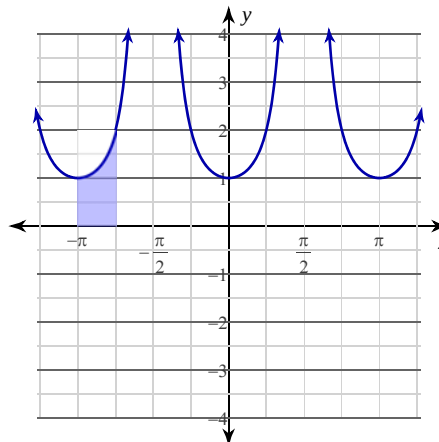
1) $y = \frac{4}{x^2}; [-2, -1]$



$$\int_{-2}^{-1} \frac{4}{x^2} dx$$

$$= 2$$

2) $y = \sec^2 x; [-\pi, -\frac{3\pi}{4}]$

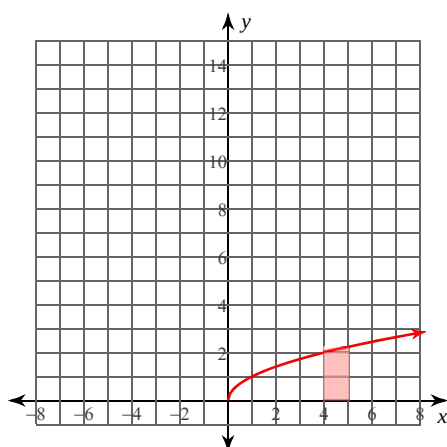


$$\int_{-\pi}^{-\frac{3\pi}{4}} \sec^2 x dx$$

$$= 1$$

For each problem, find the area under the curve over the given interval. You may use the provided graph to sketch the curve and shade the region under the curve.

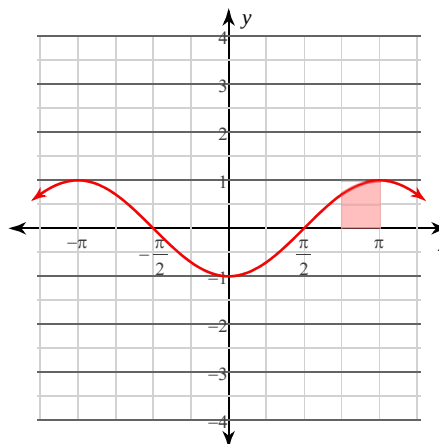
3) $y = \sqrt{x}; [4, 5]$



$$\int_4^5 \sqrt{x} dx$$

$$= \frac{2(5\sqrt{5} - 8)}{3} \approx 2.12$$

4) $y = -\cos x; [\frac{3\pi}{4}, \pi]$



$$\int_{\frac{3\pi}{4}}^{\pi} -\cos x dx$$

$$= \frac{\sqrt{2}}{2} \approx 0.707$$